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STABILITY AND EMISSIONS CHARACTERISTICS OF A LEAN PREMIXED GAS TURBINE COMBUSTOR

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The effect of incomplete fuel-air mixing on NO_x emissions and combustion stability has been studied in an atmospheric pressure, optically accessible, laboratory-scale, premixed dump combustor. The degree of fuel-air mixing was varied systematically where the fuel distribution at the combustor inlet was quantified using two-dimensional acetone fluorescence. Combustion stability was characterized both in terms of the level of combustion-generated noise during combustion oscillations and the lean blowout limit. The structure of the flame zone was characterized using two-dimensional OH fluorescence measurements that were phase locked with the combustion-generated pressure oscillations.

As has been observed previously, it was shown that incomplete fuel-air mixing results in increased NO_x emissions at overall fuel-lean conditions. The results, however, indicate that "reaction zone" NO mechanisms, enhanced by superequilibrium O-atom concentrations, are in part responsible for more than 50% of the increased NO production.

It was also shown that incomplete fuel-air mixing in a premixed combustor can significantly reduce the stable operating range of the combustor both in terms of increasing the tendency for combustion oscillation and raising the lean blowout limit. Lastly, it was observed that during unstable combustion, the flame moves in a well-defined periodic manner; however, the details of the flame behavior and instability mechanism can vary significantly depending on the overall equivalence ratio and the degree of fuel-air mixing.

Introduction

One of the more promising ultralow NO_x gas turbine combustor concepts is the lean, premixed combustor that has demonstrated the potential for significant reductions in NO_x emissions over what can be achieved by current designs [1,2]. Since it is difficult to obtain complete fuel-air mixing in actual gas turbine combustors, because of a number of constraints such as avoiding flashback and autoignition, it is important to understand the effect of incomplete fuel-air mixing on the combustor's performance. For example, studies have shown that incomplete fuel-air mixing can have a significant effect on NO_x emissions [3,4]. In addition, since premixed combustors are more susceptible to combustion instabilities [5,6], it is also important to understand the effect of incomplete fuel-air mixing on combustion stability.

The primary objective of the research reported in this paper is to characterize the effects of incomplete fuel-air mixing on both NO_x emissions and flame stability, that is, lean blowout and combustion oscillations, in lean premixed combustors. Such information is critical for the successful development of ultralow NO_x combustors. In addition, the evolution of the flame structure during unstable combustion is characterized, providing an understanding of the underlying phenomena that may lead to insights pertinent to the development of passive and active control strategies for preventing combustion instabilities.

Experimental Approach

The optically accessible dump combustor used in this study is illustrated schematically in Fig. 1. The atmospheric pressure combustor consists of a 13.7-mm-diameter by 76-cm-long mixing section, followed by a 42-mm-diameter by 38-cm-long quartz combustion section. The flame is stabilized at the step between the mixing and combustion sections, where to increase the stable operating range of the combustor a cross-hair type flame holder made of 0.89-mm-diameter stainless steel wire is located immediately downstream of the mixing section. Although actual gas turbine combustors operate at elevated pressures, a number of studies [7–9] suggest that the effect of pressure on NO_x formation is relatively weak under fuel-lean conditions, whereas near stoichiometric condition, NO_x formation has an approximately square-root pressure dependence. With this in mind, atmospheric pressure tests can

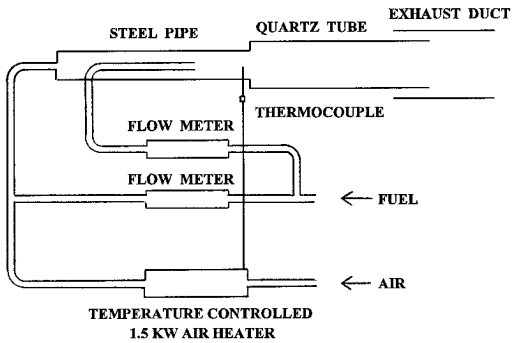


FIG. 1. Schematic drawing of dump combustor.

provide useful information regarding the behavior of lean premixed combustors.

Fuel (commercial-grade methane) is injected into the mixing section at two locations; one is 76 cm (55 mixing tube diameters) upstream of the mixing section exit, and the other is 8.2 cm (6 mixing tube diameters) upstream of the mixing section exit where the fuel is injected through a 3.4-mm-diameter co-axial fuel tube. The inlet mixture temperature is controlled at 680 ± 5 K, and the bulk velocity in the mixing section is 8.9 m/s (Reynolds number = 660). The airflow rate is 1.2 SCFM.

The degree of fuel-air mixing at the entrance to the combustion section can be varied by changing the fuel split, σ, which is defined as one minus the ratio of the mass flow rate of fuel introduced through the downstream fuel tube to the total fuel mass flow rate introduced into the combustor multiplied by 100%. It is important to note that even in the σ = 0% case, there is significant fuel-air mixing before the combustor inlet such that none of the fuel burns in a diffusion flame mode. The overall equivalence ratio (Φ_{overall}) is varied from 0.4 to 1.1, and 100%, 50%, 17%, and 0% fuel splits are investigated.

Two-dimensional acetone laser-induced fluorescence (LIF) is used to quantify the fuel distribution at the combustor inlet. A conventional gas sampling system and a chemiluminescence analyzer is used to measure the radial and axial, NO and NO_x concentration profiles in the combustor. Combustion stability is evaluated both in terms of the lean blowout limit and the occurrence of combustion oscillations, which are characterized by the intensity of the combustion-generated noise as measured using a microphone located at the combustor exit. The evolution of the flame structure during unstable combustion is characterized using two-dimensional OH LIF measurements that are synchronized with the combustion-generated pressure oscillation. Since a relatively temperature-insensitive absorption line (Q_2[8] and Q_1[9] transitions in the [1,0] vibration band of the $A^2\Sigma^+ - X^2\Pi$ system) is excited, the OH fluorescence intensity can be assumed to provide a semiquantitative indication of the OH concentration. Additional information can be found in Shih's thesis [10].

Results and Discussion

Fuel Distribution for Different Degrees of Fuel-Air Mixing

Figure 2 shows the mean and normalized standard deviation of the local equivalence ratio for two incompletely mixed cases (i.e., σ = 0% and 50%) at an overall equivalence ratio of 0.6. These results are averages from 100 individual acetone fluorescence images. The field of view is 3.5 mm by 14 mm and is located immediately downstream of the mixing section exit. At the completely mixed condition (σ = 100%), not shown in Fig. 2, the local equivalence ratio distribution is uniform and steady with a value of 0.6. For the σ = 50% case, the local mean equivalence ratio ranges from 0.4 to 0.8 with a standard deviation of up to 20%, whereas in the σ = 0% case, the local mean equivalence ratio ranges from 0.2 to 1.0 with a standard deviation of up to 40%. The fuel distribution in the incompletely mixed cases can be quantified in terms of an unmixedness parameter [11], which is defined as C'/C (1 − C) where C' is the fluctuation in the fuel concentration and C is the mean value of the fuel concentration. The average value of the unmixedness parameter over the mixing section exit area is 2.0×10^{-3} for the 0% fuel split case and 2.7×10^{-4} for the 50% fuel split case.

The Effect of Incomplete Fuel-Air Mixing on NO and NO_x Emissions

Figure 3 shows the area-weighted NO and NO_x concentrations (in ppm) as a function of the distance from the step for three different fuel-air mixing conditions at overall equivalence ratios of 0.6 and 0.92. The area-weighted concentration is the area-weighted average of the radial concentration measurements, where it is assumed that the flow field is axisymmetric. At the fuel-lean condition, for all three mixing conditions, the NO and NO_x concentrations are observed to increase with distance, which is evidence of postflame NO production. At Φ_{overall} = 0.6, it is also observed that incomplete fuel-air mixing results in increased NO production, which is consistent with previous studies [3,4]. The generally accepted explanation for increased NO production at fuel-lean, incompletely mixed conditions is in terms of thermal NO production [3,4]. However, the results shown in Fig. 3 suggest that reaction-zone NO mechanisms may contribute significantly to the total NO production.

An estimate of the reaction-zone NO can be obtained by extrapolating the NO concentrations in

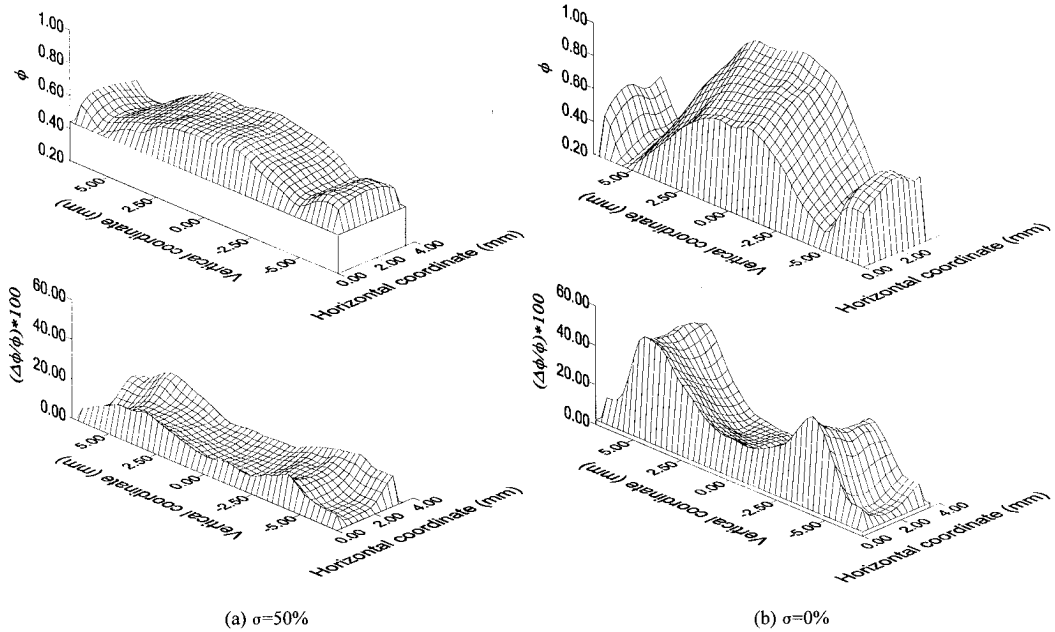


FIG. 2. Equivalence ratio distributions for different fuel splits at an overall equivalence ratio of 0.6. Vertical and horizontal coordinates correspond to radial and axial coordinates of the combustor, respectively.

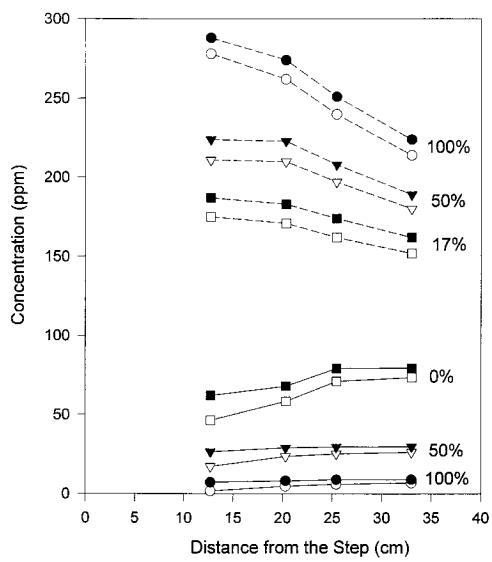


FIG. 3. Area-weighted NO and NO_x concentrations as a function of distance from step; open symbols, NO; closed symbols, NO_x; solid lines, $\Phi_{overall} = 0.6$; dashed lines, $\Phi_{overall} = 0.92$; fuel split is shown beside each curve.

Fig. 3 to the "center of gravity" of the reaction zone. The results indicate that under fuel-lean, incompletely mixed conditions, a significant amount of the NO is produced in the reaction zone (approximately 50% of the total NO in the $\sigma = 50\%$ and 0% cases), while less than 8% of the NO is produced in the reaction zone under fuel-lean, completely mixed conditions ($\sigma = 100\%$). These results indicate that the increased NO production at lean, incompletely mixed conditions cannot be explained solely in terms of the thermal NO mechanism. A possible explanation for this behavior is given later.

Model calculations have shown that in a premixed flame front, NO is formed by (1) the Zeldovich mechanism, (2) the N_2O intermediate reaction, and (3) the Fenimore mechanism, where both the Zeldovich and the N_2O intermediate mechanisms can be enhanced by superequilibrium O-atom concentrations [12]. An indication of the relative O-atom concentration can be obtained from the OH LIF measurements since one would expect partial equilibrium between OH and O [13], that is, $[O]/[O]_{equil.} = ([OH]/[OH]_{equil.})^2$.

By assuming an axisymmetric flame, a parameter, Q, can be defined as

$$Q = \frac{1}{N}\sum_{1}^{N}\int_0^r \int_0^x 2\pi\, r\, I_m^2\,(x, r)\, dx\, dr, \text{ for } I_m \geq n$$

where N is the number of the images, x and r are axial and radial coordinates, $I_m\,(x, r)$ is the OH

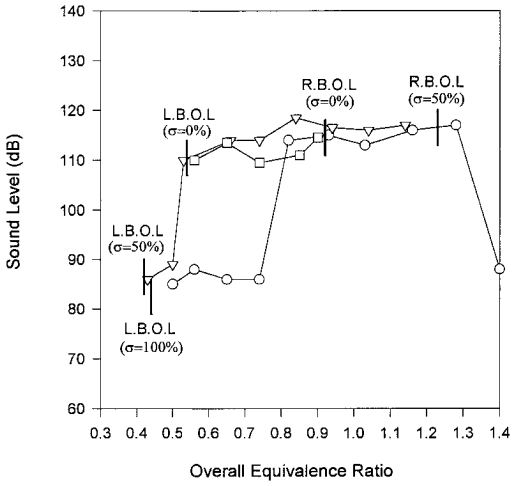


FIG. 4. Overall equivalence ratio and fuel-air mixing effects on flame stability. ○, σ = 100%; ▽, σ = 50%; □, σ = 0%.

intensity at (x, r) obtained from the two-dimensional OH LIF image and n is a selected OH fluorescence intensity threshold determined from the intensity probability distribution function. Mathematically, Q is the integration of the square of the OH fluorescence intensity over the combustion volume. It is used here as an indication of the O-atom concentration in the reaction zone. For an OH fluorescence intensity threshold of 50 (the results are relatively insensitive to the threshold value), the normalized values of Q for σ = 100%, 50%, and 0% are $Q_{100\%}/Q_{100\%} = 1.0$, $Q_{50\%}/Q_{100\%} = 2.37$, and $Q_{0\%}/Q_{100\%} = 4.08$. The increase in Q with incomplete mixing can be the result of an increased reaction zone volume and increased OH and therefore O-atom concentration, both situations leading to increased NO production. This result suggests that incomplete fuel-air mixing leads to increased superequilibrium O-atom concentration and therefore increased reaction-zone NO production in the flame zone.

Figure 3 also shows the area-weighted NO concentration as a function of distance from the step for three different fuel-air mixing conditions at an overall equivalence ratio of 0.92. It shows that, at near stoichiometric conditions, incomplete fuel-air mixing results in less NO production, which is consistent with what has been reported previously [5,6]. In the near stoichiometric case, the NO concentration decreases in the postflame region as the residence time increases. This is believed to be due to the reburning of the fuel in the postflame region [14,15], which is supported by the fact that HCN, more than 25 ppm for the σ = 100% case and 20 ppm for the σ = 50% and σ = 17% cases, was detected at the combustor exit.

The Effect of Incomplete Fuel-Air Mixing on Combustion Stability

Combustion instabilities can result in both lean blowout and combustion oscillations. The effect of incomplete fuel-air mixing on the stability characteristics of this lean premixed combustor is illustrated in Fig. 4, which is a plot of combustion-generated noise versus overall equivalence ratio for the σ = 0%, 50%, and 100% cases. In this study, when the magnitude of the combustion-generated noise was more than 20 dB above the background noise level, the combustion was considered unstable. In all cases of unstable combustion, the instability was a longitudinal instability with a single dominant frequency of ~400 Hz, which can be related to the quarter-wave natural frequency of the combustor. Also shown are the lean blowout limits for the three cases. These results show clearly that incomplete mixing results in less-stable combustion, both in terms of broadening the range over which combustion-generated oscillations occur and raising the lean blowout limit. For the completely mixed case (σ = 100%), unstable combustion occurs over the range of equivalence ratios from ~0.82 to ~1.35. In this case, the highest flame temperature occurs near the stoichiometric condition, which results in a large temperature gradient between the flame zone and the combustor wall. This large temperature gradient could induce the combustion instability in the premixed case. When Φ_{overall} is outside of this range, combustion is stable and quiet until the equivalence ratio reaches the blowout limits. That combustion becomes stable under lean and rich conditions can be attributed to the fact that the reduced heat release is not sufficient to overcome the damping in the combustor. For the σ = 50% case, unstable combustion occurs at all equivalence ratios within the blowout limits, except over a narrow range from the lean blowout limit ($\Phi_{\text{overall}} \approx 0.43$) to $\Phi_{\text{overall}} \approx 0.5$, where stable combustion is possible. For the σ = 0% case, combustion is unstable over the entire operating range of the combustor. These results indicate that the fluctuating fuel distribution under incompletely mixed conditions can contribute significantly to the heat-release fluctuations that drive unstable combustion.

The lean blowout limit is also found to be affected by incomplete fuel-air mixing. Although no significant change in the lean blowout limit was observed between the completely mixed (σ = 100%) and the σ = 50% cases, a distinct shift (from ~0.43 to ~0.55) was observed in going to the most poorly mixed case (σ = 0%).

These results on the effect of incomplete mixing on combustion stability, however, should be interpreted with care because one would expect the nature of this effect to depend on the actual shape of the fuel distribution. For example, it is possible that

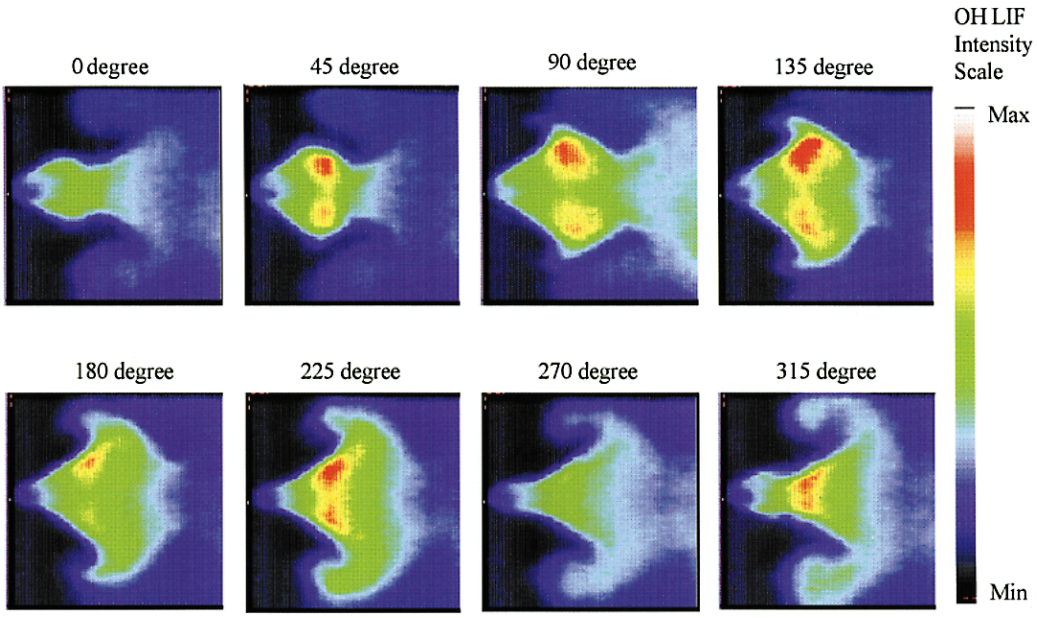


FIG. 5. Flame structure evolution for overall equivalence ratio = 0.6, σ = 50% case.

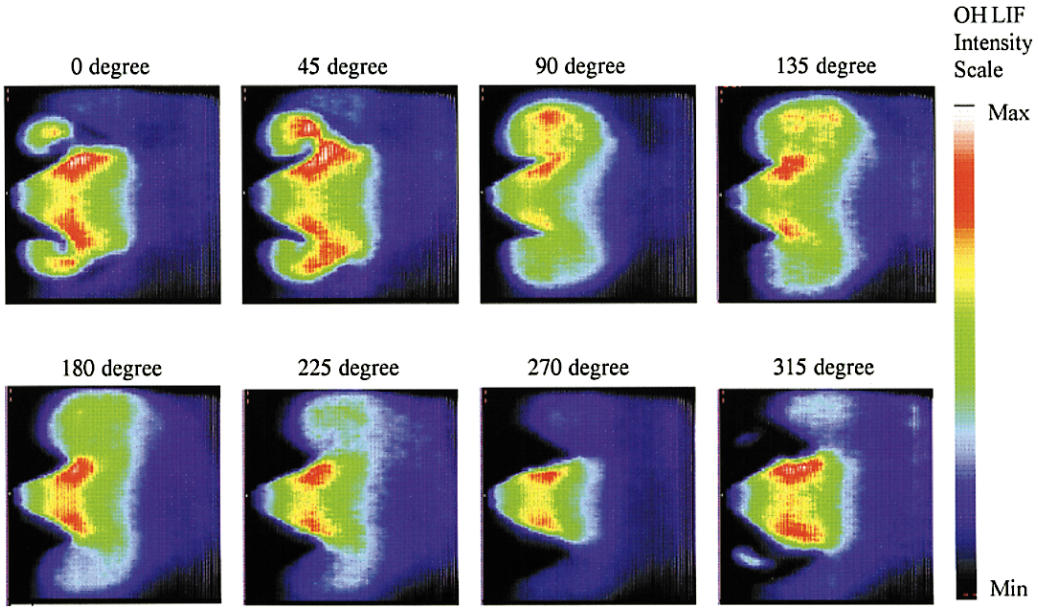
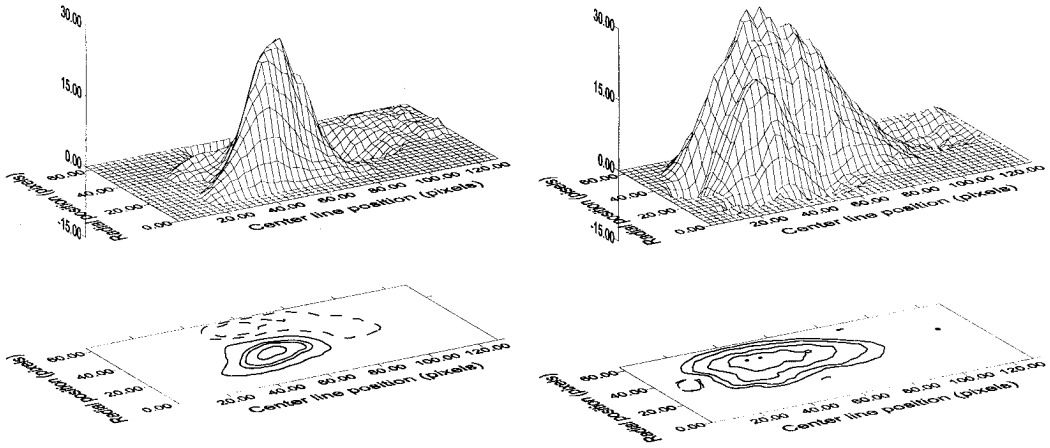


FIG. 6. Flame structure evolution for overall equivalence ratio = 0.92, σ = 100% case.



(a) $\Phi_{overall}$=0.6, σ=50% case (b) $\Phi_{overall}$=0.92, σ=100% case

FIG. 7. Local Rayleigh index distributions for different operating conditions. Solid lines, positive region; dashed lines, negative region.

certain fuel distributions could actually improve the stability characteristics of a given combustor.

Flame Structure Evolution during Combustion Oscillations

Characterization of the evolution of the flame structure during combustion oscillations can lead to useful insights regarding the phenomenology of the instability, which in turn may suggest ways to prevent its occurrence. The flame structure evolution can be reconstructed from the phase-averaged two-dimensional OH LIF measurements. Figure 5 shows the flame structure evolution during one cycle of unstable combustion at $\Phi_{\text{overall}} = 0.6$ and $\sigma = 50\%$. The field of view of each image is 35 × 35 mm and the phase angle is referenced to the sinusoidal fluctuation of the combustion-generated pressure oscillation. Note that the phase of the pressure measurement has been shifted (by ~45°) to account for the acoustic wave propagation time between the pressure measurement location and the reaction zone. At a phase angle of 0°, there is a flame pocket immediately downstream of the flame holder. The flame pocket grows and propagates downstream as the phase angle increases. At a phase angle of 90°, which corresponds to the maximum pressure, the reaction zone begins to move into the recirculation zone and continues to do so until it reaches 225°. At a phase angle of 270°, which corresponds to the minimum pressure, the reaction zone appears to be moving out of the recirculation zone and back into the flame pocket downstream of the flame holder, until the flame resumes its original shape at 0°. In this case, the instability appears to originate in the main reaction zone immediately downstream of the flame holder, which is consistent with the notion that fluctuations in the equivalence ratio at the combustor inlet are helping to drive the instability.

For the $\Phi_{\text{overall}} = 0.92$ and $\sigma = 100\%$ case, the flame structure evolution illustrated in Fig. 6 is observed to be quite different from that discussed earlier. At a phase angle of 270°, there is a main reaction zone downstream of the flame holder (similar to the $\Phi_{\text{overall}} = 0.6$, $\sigma = 50\%$ case at a phase angle of 0°). At a phase angle of 315°, small pockets of flame are observed in the recirculation zone. These flame pockets then propagate and connect with the main reaction zone at 45°. At a phase angle of 90°, winglike structures that gradually decrease in size as the phase angle increases appear beside the main reaction zone. In this case, the instability appears to originate in the wall region and recirculation zone, while the main reaction zone, that is, immediately downstream of the flame holder, remains relatively unchanged.

Another way to analyze these measurements is in terms of the Rayleigh criterion [16], which states that when the unsteady heat release $q'(\xi, t)$ is in phase with the pressure fluctuation $p'(\xi, t)$, the pressure wave oscillation can be sustained. This criterion can be expressed in terms of the Rayleigh index, R, as follows:

$$R = \frac{1}{V}\frac{1}{T}\int_V \int_T p'(\xi, t)\, q'(\xi, t)\, dt d\xi > 0$$

where ξ is a spatial coordinate, t is time, T is the

instability period, and V is the volume of the flame. Using the fluctuating OH fluorescence intensity as an indicator of heat-release fluctuation along with the measured combustion-generated pressure fluctuation, the local and global Rayleigh indices can be calculated from the two-dimensional in-phase OH LIF images. In calculating the local and global Rayleigh indices, the pressure is assumed constant in phase and amplitude throughout the entire flame-zone, which is reasonable since the flame is concentrated near the flame holder and is small compared to the acoustic wavelength.

Figure 7 shows the local Rayleigh index over the upper half of the combustion section for the two unstable cases discussed previously, indicating that even when the overall combustion process is unstable, regions where the instability is driven ($R > 0$) and regions where it is damped ($R < 0$) exist. For the $\Phi_{\text{overall}} = 0.6$, $\sigma = 50\%$ case, the Rayleigh index is positive in the central flame pocket region where the heat release caused by the pulsating flame is in phase with the pressure oscillation. Damping of the instability happens near the wall region where the heat release in the recirculation zone is out of phase with the pressure oscillation. In the $\Phi_{\text{overall}} = 0.92$, $\sigma = 100\%$ case, as shown in Fig. 7b, the instability is driven near the wall region because the heat release from the winglike structures is in phase with the pressure oscillation. In this case, there is only a small region in the recirculation zone where damping occurs.

Global Rayleigh indices for the instability-driving regions and for the damping regions are obtained by integrating the local Rayleigh index over the respective regions. The global Rayleigh indices for the driving and damping regions, respectively, are $+0.63$ and -0.37 for the $\Phi_{\text{overall}} = 0.6$, $\sigma = 50\%$ case and $+0.92$ and -0.08 for the $\Phi_{\text{overall}} = 0.92$, $\sigma = 100\%$ case. For the $\Phi_{\text{overall}} = 0.92$, $\sigma = 100\%$ case, the instability is driven over a significant portion of the flame zone.

Recently, the use of active control to prevent combustion oscillations has been demonstrated in afterburners and dump combustors [17–19], where modulated secondary fuel injection strategies have been employed to effectively cancel out the heat-release fluctuations. Information such as that shown in Fig. 7 would be valuable in defining the optimum location for secondary fuel injection in an active control system.

Conclusion

A study of the effect of incomplete fuel-air mixing on NO_x emissions and flame stability in a laboratory scale premixed combustor is presented. As has been observed previously, it was shown that incomplete fuel-air mixing results in increased NO_x emissions at overall fuel-lean conditions. The results, however, indicate that reaction-zone NO mechanisms, enhanced by superequilibrium O-atom concentrations, are in part responsible for more than 50% of the increased NO production.

It was also shown that incomplete fuel-air mixing in a premixed combustor can significantly reduce the stable operating range of the combustor both in terms of increasing the tendency for combustion oscillation and raising the lean blowout limit. Finally, it was observed that during unstable combustion, the flame moves in a well-defined periodic manner; however, the details of the flame behavior and the instability mechanism can vary significantly depending on the overall equivalence ratio and the degree of fuel-air mixing.

Acknowledgments

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